

ABHIJEET TIWARI

Senior Agentic AI Forward Deployed Engineer · GenAI Solutions Architect · 10+ Years

Agentic AI · Multi-Agent Systems · Forward Deployed Engineering · RAG & LLMOps · Evals & Harness Engineering · MCP · Solution Architecture · Full-Stack

Pune, India

PROFESSIONAL SUMMARY

Senior Agentic AI Forward Deployed Engineer and GenAI Solutions Architect with 10+ years of shipping production software, now embedded with enterprise customers to take agentic AI from whiteboard to production. Designs and builds multi-agent systems end to end, including orchestration graphs, tool and function-calling, RAG and real-time data pipelines, structured outputs, evaluation harnesses, agent observability and tracing, LLMOps, and safety guardrails, all on a deep foundation of Python, TypeScript, Node.js/NestJS, React/Next.js, Nuxt/Vue 3 and .NET across AWS and GCP.

Operates as the technical bridge between the business and the build: sits with operators, product owners and domain experts to map how work actually flows, co-defines the agent workflows, decision points, hand-offs and human-approval gates with the client team, then owns delivery through integration, evaluation and adoption. Track record spans supply chain (SAP S/4HANA, ServiceNow), ESG/utility data, revenue and sales operations, HR/PEO services, life-sciences logistics and large-scale legacy modernization.

Certified by Anthropic as a Claude Certified Architect (Foundations level); fluent across Claude, Gemini and OpenAI/Codex and their agent-skill ecosystems, and comfortable running the same agent harnesses on OpenCode and local models via Ollama; builder of AI-native tooling including a custom Model Context Protocol (MCP) suite that gives agents persistent memory. Proven technical lead and mentor who ships measurable outcomes, not demos.

SIGNATURE OUTCOMES

- **Modernized ~707K LOC of legacy code across 3 core modules in ~12 weeks** with an AI-native, 18-node multi-agent SDLC harness: ~250 user features, 237 REST endpoints, 168 screens, 3,439 automated tests, ~1.2 bugs per KLOC.
- **Architected a 6-agent GCP supply-chain platform for Waymo** on Google ADK and Vertex AI Agent Engine, grounded in BigQuery and integrated with SAP S/4HANA and ServiceNow.
- **Shipped production agentic document extraction (Measurabl BillAI)** with schema-validated outputs, confidence gating, human-in-the-loop review and a Langfuse eval loop.
- **Compressed 13.9M tokens of engineering context to ~1M (14:1)** in a persistent-memory MCP layer shared across coding agents (Claude Code, Codex, OpenCode).

CORE TECHNICAL SKILLS

Agentic AI & Forward Deployment: Multi-agent orchestration, agent workflow design with business stakeholders, tool/function-calling, MCP servers & clients, agent skills, human-in-the-loop & approval gates, prompt-injection guardrails, structured outputs (JSON Schema), long-term agent memory, context engineering, harness engineering

LLM Engineering & LLMOps: RAG (chunking, embeddings, hybrid retrieval, grounding), prompt engineering, evaluation harnesses & regression evals, agent observability/tracing, Langfuse, per-task model routing (capability vs. cost vs. latency), inference/cost optimization, OCR & document extraction, fine-tuning fundamentals

Models, Frameworks & Tooling: Coding-agent runtimes (Claude Code, OpenAI / Codex, OpenCode, Ollama, Cursor), Anthropic Claude (Agent SDK, Claude Skills), Google Gemini, Google ADK, Vertex AI Agent Engine & RAG Engine, LangChain, LangGraph, LlamaIndex, Model Context Protocol (custom + built-in servers)

Architecture & Solutioning: Solution architecture, system design, event-driven & microservice architecture, Clean Architecture, API design (REST, GraphQL, gRPC), UML/sequence diagrams, Miro, Excalidraw, technical proposals, SOWs & client presentations

Languages: Python, TypeScript, JavaScript, C# (.NET 10), GoLang, SQL, VB.NET (legacy reading), HTML, SCSS

Backend & Data: Node.js, NestJS, .NET 10 / EF Core, Python services, Prisma, PostgreSQL, MySQL, SQL Server, MongoDB, BigQuery, stored-procedure integration, webhooks, async job pipelines

Frontend: React.js, Next.js, Nuxt 4 / Vue 3, Element Plus, RemixJS, React Native, Redux Toolkit, Pinia, TailwindCSS

Cloud & DevOps: AWS (Lambda, S3, EC2, RDS, Cognito), GCP (Vertex AI, BigQuery), Azure, Terraform (IaC), Docker, GitHub Actions, CI/CD quality gates, Vercel

Enterprise Integrations: SAP S/4HANA, ServiceNow, Salesforce/CRM, Jira, Confluence, GitHub, Slack, Figma

Testing & Security: Playwright (incl. AI-generated e2e), Jest, Cypress, React Testing Library, Snyk, CodeQL, JWT, OAuth 2.0, row-level security preservation

PROJECT EXPERIENCE

Simplifi: AI-Native Legacy Modernization Platform (R Systems)

Senior Agentic AI Forward Deployed Engineer / Modernization Architect

Enterprise modernization program that migrated a large legacy agency-management product (Aurelia + ASP.NET WebForms + VB.NET + SQL Server) to a .NET 10 Clean Architecture backend and a Nuxt 4 / Vue 3 frontend, delivered by a small team amplified by a custom multi-agent SDLC harness.

- Modernized ~707,000 LOC / 2,208 files across three core modules (Job Jacket, Estimates, Timesheets) in ~12 weeks (~4 weeks per module including discovery): ~250 end-to-end user features, 237 REST endpoints across 31 controllers and 168 Nuxt screens, backed by 3,439 automated tests (28% test-to-prod LOC ratio) and a ~1.2 bugs/KLOC defect rate.
- Designed and built an AI-native SDLC “factory”: an 18-node, 5-loop agent graph (intent → discovery → contract → plan → codegen → parity inspection → self-review → adversarial review → QA → draft PR) with role-based model routing (cheap models for extraction, frontier models only at the gates), per-run git worktrees, bounded retry/escalation loops and a critic-never-below-producer invariant.
- Authored a 14-skill agent harness with 25+ specialized sub-agents, orchestrated as loops and graphs that are agnostic to which coding agent executes them (runs unchanged on Claude Code, Codex or OpenCode): legacy module discovery (4 parallel layer agents), PM and architect review, UAT parity-scenario generation with an independent coverage auditor, an e2e bug-hunt loop that runs scenarios via Playwright MCP and traces failures to file and line, flow-parity review, code review, pixel-parity checks (numeric $\Delta E/px$ diffs vs. Figma) and a Slack-to-Jira CS bug intake pipeline.
- Built a Model Context Protocol layer that blends custom-built servers (module discovery, read-only DB inspection, persistent “hivemind” memory) with built-in / off-the-shelf MCP servers (GitHub, Atlassian, Playwright, Figma, Slack) behind one interface, compressing 13.9M tokens of discovery context to ~1M (14:1) in a shared store so every module started warmer than the last, regardless of which coding agent picked it up.
- Worked directly with the client's product owners, QA and CS teams to define which workflows the agents own (discovery, scenario generation, bug triage, draft PRs) and where humans stay in the loop (architecture review, PR approval, scenario sign-off); made legacy code the arbiter for every behavioural decision so parity disputes were settled with evidence, not opinion.

- Preserved business-critical rules by reusing existing stored procedures and keeping row-level security (SEC_CLIENT / EMP_OFFICE / CP_SEC_CLIENT) inside the database layer; layered per-PR quality gates (Danger, automated review, security agent, build/test/format, CODEOWNERS) with draft-PR-only, protected-branch guardrails.
- Shipped a containerized headless agent (Docker plus a configurable coding-agent runtime) that takes a CS bug report from a Slack thread through Jira creation, staging reproduction, cross-repo RCA and a draft PR, with three guardrail layers and zero database write access.

Tech: Agent orchestration graphs and loops, Claude Code, Codex, OpenCode, MCP (custom + built-in servers), Playwright MCP, .NET 10 / C#, EF Core, SQL Server, Nuxt 4 / Vue 3, Element Plus, Pinia, TypeScript, Docker, GitHub Actions, Jira, Slack, Figma

Waymo: SAP Supply Chain Multi-Agent POC **Lead GenAI Architect / Forward Deployed Engineer**

GCP-native multi-agent platform for end-to-end supply-chain operations (planning, procurement, inventory, logistics, service) at Waymo, integrated with SAP S/4HANA and ServiceNow.

- Architected the platform on Google's Agent Development Kit (ADK) and Vertex AI Agent Engine: six specialized domain agents orchestrated by a root LlmAgent with explicit routing, tool/function-calling contracts and shared session state.
- Designed retrieval and grounding with the Vertex AI RAG Engine over enterprise data in BigQuery, and integrated SAP S/4HANA and ServiceNow for live supply-chain context and closed-loop actions (ticketing, status updates).
- Selected and tuned Gemini 2.0 Flash / Pro per agent to balance latency, cost and capability across planning, retrieval and execution tasks; defined guardrails, human-approval checkpoints for write actions and the evaluation strategy for agent decisions.
- Ran discovery workshops with supply-chain operators and SAP functional leads to map real operating procedures into agent workflows, clarifying which agent owns which step, what data each needs, and where a human must confirm, then led solutioning and architecture reviews with client stakeholders through to POC sign-off.

Tech: Google ADK, Vertex AI Agent Engine, Gemini 2.0 Flash/Pro, Vertex AI RAG Engine, BigQuery, SAP S/4HANA, ServiceNow, multi-agent orchestration, Python, GCP

Measurabl: BillAI (Agentic Utility-Bill Intelligence) **GenAI Architect & Lead / Forward Deployed Engineer**

Measurabl is the most widely adopted ESG data platform for commercial real estate. BillAI is its AI-powered utility-bill ingestion engine, integrated with the WegoWise utility-data product, that turns unstructured utility statements into structured, analyzable data.

- Architected an end-to-end agentic RAG pipeline that ingests scanned and PDF utility statements, performs OCR, and applies retrieval-augmented LLM extraction to produce schema-validated structured data, engineered for high field-level accuracy with explicit hallucination controls.
- Designed an asynchronous, event-driven architecture: a Rails backend orchestrates extraction requests against a NestJS BillAI API, dispatches work to LLM extraction workers and receives results through webhook callbacks, with state and raw data persisted in MySQL.
- Defined the prompt architecture and JSON-Schema structured outputs, adding confidence scoring and a human-in-the-loop review queue that gates low-confidence results before they reach downstream ESG reporting.
- Built the evaluation and observability harness on Langfuse to measure extraction quality, prompt/model regressions, latency and token cost across versions, forming a data-driven loop for safe prompt and model iteration in production.
- Partnered with Measurabl's data-operations and product teams to define the review workflow, confidence thresholds and exception paths, and to align extraction fields with the WegoWise data model; developed the operator/review UI in Next.js and provisioned the AWS footprint as code with Terraform.

Tech: Agentic RAG, LLM extraction, OCR, prompt engineering, structured outputs, Langfuse (evals & observability), Rails, NestJS, Next.js, MySQL, AWS, Terraform, TypeScript

OptimaAI: Agentic SDLC Platform (R Systems / Exiqo Labs)

GenAI / Platform Architect

OptimaAI is a multi-agent platform that automates the software development lifecycle, turning product intent into engineering artifacts and accelerating delivery across teams.

- Architected a system of specialized, collaborating agents: a PRD-to-Jira agent that decomposes requirement documents into structured epics, stories and acceptance criteria, and a Code-to-Documentation / Discovery agent that generates technical docs and codebase walkthroughs from source repositories.
- Designed the orchestration layer, covering agent routing, tool/function-calling and shared long-term memory, and extended it to automated test-case generation, PR review and summarization, release-note generation, requirement-to-ticket traceability and knowledge-base Q&A over Confluence and GitHub.
- Engineered safety and governance with prompt-injection guardrails, input/output validation and human-approval checkpoints, keeping autonomous actions safe, auditable and reversible; defined a per-task model strategy balancing capability against cost.
- Worked with delivery leads and product managers across engagements to map their real SDLC hand-offs into agent workflows, pilot them on live projects, and iterate on prompts and evals from practitioner feedback until output quality met team standards.

Tech: Multi-agent orchestration, LangGraph, Claude / LLM APIs, function-calling, RAG, prompt-injection guardrails, Jira / Confluence / GitHub APIs, TypeScript, Python

G&A Partners: Agentic AI Sales Assistant

GenAI Engineer / Architect / Forward Deployed Engineer

G&A Partners is a leading HR outsourcing / PEO provider. Built an agentic AI sales assistant that automates the sales workflow end-to-end for its revenue team.

- Built an agentic sales assistant that automates lead qualification, personalized outreach, follow-ups and proposal drafting, grounded in product, pricing and CRM data via retrieval.
- Designed multi-step agent workflows with tool/function-calling into CRM and communication systems, with guardrails and human approval gating every outbound action.
- Instrumented the agent with evaluations and tracing to track response quality, grounding faithfulness and safe-action compliance, giving sales leadership transparency into every agent decision.
- Shadowed sales reps and worked with sales-ops leadership to define qualification criteria, tone/compliance rules and hand-off points, then translated them into agent playbooks and eval rubrics the business could own and update.

Tech: Claude / LLM APIs, agentic workflows, RAG, CRM integrations, prompt engineering, evals & tracing, TypeScript, Python

TAL: Agentic FSD & Document Intelligence Platform

GenAI Architect / Lead

Agentic document-intelligence system that turns legacy functional specifications, SOWs and change requests into new, traceable engineering artifacts, validated on real legacy warehouse-management (Exacta WMS) specifications for a life-sciences logistics client.

- Architected and built an agentic RAG workflow over enterprise documents: parsing, chunking, embedding and indexing legacy FSDs into a vector store, then orchestrating multi-step retrieval and LLM reasoning to answer questions and act on document content.
- Designed an agent pipeline that reads a change order plus the legacy FSD corpus and generates approved-quality deliverables: user stories, a full Functional Specification Document (architecture, module interactions, configuration reference, error handling, backward-compatibility analysis), QC test cases and a user-story-to-FSD traceability matrix.
- Engineered retrieval quality and grounding with citation-or-UNVERIFIED rules and guardrails, so every generated requirement traces back to source documents with no invented behaviour.

- Worked with the client's business analysts and QC leads to define the artifact templates, approval workflow and review checkpoints, so generated FSDs slot directly into their existing change-order process.

Tech: Agentic RAG, document ingestion (parsing / chunking / embedding), vector store, structured generation, traceability, LLMs, Python, NestJS, AWS

Custom MCP Suite: AI-Native Tooling with Persistent Memory

Architect, Creator & Author

Self-initiated R&D, published on LinkedIn, that gives AI agents persistent memory through the Model Context Protocol (MCP).

- Designed and built a custom MCP server suite giving AI agents persistent, queryable long-term memory across sessions and tools: memory schema, storage and retrieval interfaces, and the exposed tool surface, later productionized as the “hivemind” layer in the Simplifi modernization program.
- Implemented MCP tools and resources that expose memory read/write to any MCP-compatible client (Claude Code, Codex, OpenCode, Cursor), enabling context continuity across conversations and agents; authored the LinkedIn article “How I Built My Own Custom MCP Suite to Give AI Agents Memory.”

Tech: Model Context Protocol (MCP), TypeScript, Python, SQLite / vector memory store, retrieval, Claude Code, Codex, OpenCode

QA Agent: Automated Playwright Test Generation

GenAI Engineer / Architect

An AI agent that automatically authors end-to-end test coverage, generating Playwright scripts and structured test cases from application context.

- Designed and built an AI QA agent that generates Playwright end-to-end scripts and structured test cases from requirements, user stories and UI/DOM context, exploring application flows to derive happy-path, edge and negative scenarios wired into CI.
- Added evaluation and self-correction loops that validate and repair generated tests against the live application before commit; co-defined coverage priorities and review gates with QA leads so the agent complemented, rather than bypassed, their process.

Tech: Claude / LLM APIs, agentic workflows, Playwright, TypeScript, CI/CD, evals

VTS: Visitor & Tenant Experience Platform

Solutions Architect / Full-Stack Lead

Commercial real-estate platform streamlining building access and on-site experience for property managers and tenants at scale.

- Architected and delivered multi-day passes, additional-recipient notifications and streamlined visitor flows on an event-driven microservices architecture (TypeScript/React, GoLang and Node.js services, GraphQL, gRPC); owned the AWS cloud architecture via Terraform.
- Introduced AI-driven enhancements, including intelligent notification routing and anomaly detection across visitor and access patterns, to improve security and operational efficiency.

Tech: TypeScript, React.js, GoLang, Node.js, GraphQL, gRPC, PostgreSQL, AWS, Terraform

Inbillo: Contract Intelligence Platform

Full-Stack Lead / Architect

Contract-management platform helping companies manage financial liquidity by browsing, searching, storing and transacting against contracts, with third-party and cryptocurrency payments.

- Architected the platform (listings, search/filtering, invoicing, checkout, admin bulk upload) and introduced LLM-based contract understanding, extraction and semantic search to accelerate retrieval and reduce manual review.
- Integrated MetaMask crypto payments and third-party shipping/payment APIs; owned the React/Redux frontend, CI/CD and Vercel deployments.

Tech: React.js, Node.js, Express.js, Redux, MetaMask, TypeScript, Vercel, LLM document extraction, semantic search

TapGoods: Rental Marketplace & Inventory Platform

Full-Stack Lead / Architect

Inventory-management and marketplace software for rental businesses spanning catalog, operations and a customer-facing storefront.

- Architected the Next.js / NestJS application with real-time collaboration over WebSockets, OAuth-secured services, GraphQL caching and query optimization on AWS (Cognito, S3, EC2).
- Introduced AI-driven search and recommendations across the rental catalog to improve discovery and conversion.

Tech: Next.js, NestJS, TailwindCSS, Socket.IO, GraphQL, AWS

NeuronX.ai: Data Management & Analytics Platform

Full-Stack Developer / Architect

Data-management and analytics platform running enrollment and competitions and surfacing performance insights through dashboards.

- Architected enrollment and competition-management interfaces with role-based admin permissions on AWS Lambda and Prisma; built ML-driven performance-analytics dashboards.
- Remediated the majority of security vulnerabilities across portal and admin app, automated DB deployments via GitHub Actions and added Jest / RTL coverage.

Tech: React.js, Node.js, Prisma, AWS Lambda, analytics & ML-driven insights

BioMark: AI-Assisted Patient & Doctor Portal

Lead Frontend Developer / Architect

Healthcare platform offering patient and doctor portals across web and mobile for managing disease-wise medical data and coordinating care.

- Architected patient and doctor portals (web + mobile) with disease-wise data management and role-based permissions; built AI-assisted patient triage that prioritizes and escalates cases by urgency, plus real-time chat and appointment scheduling.
- Designed the data model and access controls for sensitive medical records with privacy and role separation in mind.

Tech: React.js, Redux, TypeScript, SCSS, Node.js, Socket.IO, Azure

CERTIFICATIONS

Anthropic / Claude

- Claude Certified Architect (Foundations), Anthropic (May 2026) · Credential ID: bdovsipqn6o9
- Claude 101, Anthropic (May 2026) · Credential ID: rmczzmztgcao
- Building with the Claude API, Anthropic (Mar 2026) · Credential ID: cev27upkga4
- Claude Code in Action, Anthropic (Mar 2026) · Credential ID: k9rzhxrr7f56
- Introduction to Model Context Protocol (MCP), Anthropic (Mar 2026) · Credential ID: njwa5sqk5gcx
- Introduction to Agent Skills, Anthropic (Mar 2026) · Credential ID: iuc3w9xdon8e

Machine Learning, Data Science & Other

- Developing Machine Learning Applications, AWS Training & Certification (2020)
- Python for Data Science, IBM (Cognitive Class)
- Geoprocessing using Python, Indian Institute of Remote Sensing (IIRS), ISRO (2021)

RECENT AWARDS & ACHIEVEMENTS

- Spot Award Winner (2025)
- Solution Enabler (2025)
- Innovation Award, Individual Excellence (2025)